

Notes: Huberman and Regev (JF, 2001)

Contagious Speculation and a Cure for Cancer: A Nonevent that Made Stock Prices Soar

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1 Big picture

- **Question.** Can media publicity move stock prices even when no genuinely new information is released?
- **Setting.** The paper studies EntreMed (ENMD), a small biotechnology firm linked to a potential anti-cancer breakthrough. The same underlying scientific result appeared in *Nature* and in mainstream news in late November 1997, but ENMD's largest price move occurred only after a dramatic, front-page *New York Times* story on Sunday, May 3, 1998.
- **Core challenge.** In most event studies, media coverage and information arrival are bundled together. Here the authors exploit a rare case in which the later media event looked like news but was not actually new.
- **Main strategy.** Compare price and volume reactions across three dates:
 - November 28, 1997: real positive news,
 - May 4, 1998: no-new-news publicity,
 - November 12, 1998: real negative news.

The paper then checks whether the May 4 enthusiasm spilled over to other biotechnology firms.

- **Main result for ENMD.** ENMD closed at \$12.063 on Friday, May 1, 1998, opened at \$85 on Monday, May 4, and closed at \$51.81. The Friday-close-to-Monday-close return was about **330%**. A large part of this move was permanent: the stock ended that week at \$33.25 and stayed well above its pre-May level for months.
- **Main cross-stock result.** The optimism was contagious. The equally weighted Nasdaq Biotechnology Combined Index rose **7.5%** on May 4, excluding ENMD. Seven biotech stocks rose more than **25%**, with some jumping over **100%**, despite no obvious firm-specific news.
- **Key comparison.** When the scientific breakthrough actually became public in November 1997, ENMD rose only **28.4%**. When genuinely negative replication news arrived on November 12, 1998, ENMD fell about **24%**. So the biggest and most persistent reaction came on the *nonevent* day.
- **Why this matters.** The paper is not just saying markets sometimes overreact. Its stronger claim is that *public attention itself* can create a large and lasting repricing, even when the underlying information set has not changed.
- **Economic implication.** If prices react to salience and publicity rather than only to fundamentals, then capital allocation can become partly arbitrary. A firm's financing conditions may improve or deteriorate because of attention shocks rather than because of genuine changes in expected cash flows.
- **Bottom line.** This is one of the cleanest early papers showing that publicity and investor attention can move prices in a way that standard efficient-markets reasoning has trouble rationalizing.

2 Data and measurement (key objects)

2.1 Primary data sources

- **Intraday data.** NYSE Trade and Quotes (TAQ) for transaction prices, quotes, and trading volume.
- **Daily return data.** CRSP through December 1997.
- **Later daily data.** CSI / Money Central Investor Web Site for 1998–1999 observations.
- **Firm-specific institutional details.** SEC filings for shares outstanding and ownership around May 1998.
- **Industry membership.** Nasdaq’s Biotechnology Combined Index and its membership rules.

2.2 Unit of observation and event windows

- **Single-firm evidence.** ENMD stock prices and volumes around key event days.
- **Sector evidence.** Daily returns and trading volume for firms in the Nasdaq Biotechnology Combined Index, excluding ENMD when measuring contagion.
- **Large-firm spillover evidence.** Bristol-Myers Squibb (BMY), which had a strategic partnership with ENMD.
- **Event dates.**
 - November 27–28, 1997: *Nature* publication and immediate press coverage.
 - May 3–4, 1998: prominent *New York Times* front-page article and next-day market reaction.
 - November 12, 1998: *Wall Street Journal* report that other labs failed to replicate the results.
 - February 10, 1999: later ENMD–BMY agreement modification used as an additional comparison point for BMY.

2.3 Key empirical objects

The paper is mostly forensic event analysis rather than a regression paper, so the essential objects are simple return and volume measures.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \quad (1)$$

- **Interpretation.** Daily return is the basic measure of market reaction on each event date.

For BMY, the paper also reports excess return relative to the NYSE:

$$ER_t^{\text{BMY}} = R_t^{\text{BMY}} - R_t^{\text{NYSE}}. \quad (2)$$

- **Interpretation.** This strips out broad market movements and asks whether BMY moved unusually even after controlling for the market’s return that day.

For volume comparisons, the paper uses both raw share volume and relative volume:

$$RV_t^{\text{BMV}} = \frac{\text{Volume}_t^{\text{BMV}}}{\text{Volume}_t^{\text{NYSE}}}. \quad (3)$$

- **Interpretation.** This asks whether trading intensity in BMV was unusually high relative to aggregate market activity.

For the biotechnology sector, the equally weighted index return is conceptually

$$R_t^{\text{EW}} = \frac{1}{N} \sum_{i=1}^N R_{it}, \quad (4)$$

while the value-weighted return is

$$R_t^{\text{VW}} = \sum_{i=1}^N w_{i,t-1} R_{it}. \quad (5)$$

- **Interpretation.** The equally weighted return captures how typical firms in the biotech sector reacted. The value-weighted return shows whether the reaction also mattered in dollar terms among larger firms.

The paper also reports ENMD’s annualized volatility between late November 1997 and early May 1998 as

$$\sigma_{\text{ann}} = \left(254 \cdot \frac{\sum_t R_t^2}{T} \right)^{1/2}. \quad (6)$$

- **Interpretation.** Even before the May 1998 surge, ENMD was a highly volatile biotechnology stock. This matters because the paper is not claiming the stock was normally quiet; it is claiming the May 4 move was extraordinary *even relative to that volatile environment*.

2.4 Observed versus unobserved variables

- **Observed.** Prices, returns, trading volume, publication timing, media prominence, and cross-stock comovement within biotech.
- **Unobserved.** Investor beliefs, attention, interpretation of the breakthrough, and expected future cash flows.
- **Why this is enough for the paper’s purpose.** The paper does not need a structural belief model. Its point is comparative: the market reacted much more to old information repackaged with salience than to the original information when it first arrived.

3 Models and empirical specifications (all equations + interpretation)

3.1 This is not a structural or regression paper

- There is **no formal equilibrium model** and **no multivariate return regression** in the paper.

- The method is a tightly argued event-study-style comparison built around a rare chronology:
 - same underlying scientific result,
 - different degrees of publicity,
 - different price and volume responses.
- The paper’s “identification” is therefore narrative and comparative rather than econometric.

A concise way to summarize the paper’s logic is the diagnostic inequality

$$|\Delta P_{\text{May 4, 1998}}^{\text{nonevent}}| \gg |\Delta P_{\text{Nov. 28, 1997}}^{\text{true news}}|, \quad (7)$$

with

$$|\Delta P_{\text{May 4, 1998}}^{\text{nonevent}}| > |\Delta P_{\text{Nov. 12, 1998}}^{\text{bad news}}|. \quad (8)$$

- **Interpretation.** If the efficient-markets benchmark were working in the usual way, the response to genuine information should dominate the response to pure publicity. In the data, the opposite happens.

3.2 ENMD around the three key dates

- **November 28, 1997: real good news.**
 - The scientific result appeared in *Nature* on November 27, 1997.
 - The *New York Times*, Newsday, CNBC, CNN, and ENMD’s own press release all covered the development immediately.
 - ENMD’s closing price rose from \$11.875 on November 26 to \$15.25 on November 28.
 - This is a **28.4%** increase.
 - Volume on November 28 and December 1 was unusually high, which suggests that the market did in fact notice the news.
- **What an EMH-style reading would say.** The market got the real news in late November 1997, priced it, and then moved on.
- **May 4, 1998: no-new-news publicity shock.**
 - The Sunday, May 3 *New York Times* article appeared on the **front page**, upper left corner, with the label “**A special report.**”
 - The article repeated the same core scientific development from November 1997 but wrapped it in exceptionally vivid and optimistic language.
 - ENMD had closed at \$12.063 on Friday, May 1.
 - On Monday, May 4, it opened at \$85 and closed at \$51.81.
 - The Friday-close-to-Monday-close return was about **330%**.
- **Persistence of the nonevent effect.**
 - ENMD closed the week at \$33.25, still almost three times the prior Friday’s close.
 - It did not close below \$20 until late August 1998.

- By late fall it still had not closed below \$16.94, about **40%** above its May 1 price.
- **November 12, 1998: real bad news.**
 - The *Wall Street Journal* reported that other labs failed to replicate the result.
 - ENMD fell from \$32.625 on November 11 to \$24.875 on November 12, about a **24%** drop.
 - Even after this truly negative news, ENMD still traded at more than twice its May 1 pre-publicity level.

Interpretation.

- The core contrast is stark: the market responded moderately to real scientific news, massively to recycled front-page publicity, and materially but less dramatically to later bad scientific news.
- That ranking is the opposite of what a simple “prices move only on new information” story would lead one to expect.

3.3 Cross-stock contagion in the biotech sector

The paper asks whether the May 4 reaction was purely idiosyncratic to ENMD or whether the publicity shock spilled over to related firms.

- The Nasdaq Biotechnology Combined Index (excluding ENMD) had **134** member firms.
- On May 4, 1998, the **equally weighted** index returned **7.5%** and the median stock rose **1.6%**.
- This was highly unusual: in 1996–1997, the absolute value of **506 of 507** daily returns of this equally weighted portfolio was below **6%**.
- The **value-weighted** index return was **1.43%**, versus only **0.29%** for Nasdaq on that day.

The paper then isolates the seven biotech stocks with May 4 returns above 25%.

- Three of these rose more than **100%**.
- Two rose between **50% and 100%**.
- Two rose between **25% and 50%**.
- A search in ABI Inform finds **no mention** of these firms from May 1 through May 4.

The paper also forms a value-weighted portfolio of these seven extreme winners.

$$R_{7, \text{May 4}}^{VW} \approx 75\%. \quad (9)$$

Over the subsequent horizons:

$$R_{7, \text{week from May 1}}^{VW} = 35\%, \quad R_{7, \text{two weeks from May 1}}^{VW} = 27\%. \quad (10)$$

- **Interpretation.** More than one-third of the initial **75%** jump remained after two weeks.
- In dollar terms, these seven stocks gained about **\$185 million** on May 4, and about **\$66 million** of that increase was still there on May 15.

Why this matters.

- A purely rational-information story would predict that the main sector response should have occurred in late November 1997, when the *Nature* article actually appeared.
- Instead, the big spillover appears in May 1998, when the informational content was essentially unchanged but the publicity was much more dramatic.

3.4 Bristol-Myers Squibb as a large-firm spillover test

BMY is useful because it is a huge, visible firm and a plausible commercial beneficiary if ENMD's breakthrough mattered.

- The November 27, 1997 and May 3, 1998 *Times* articles both mention BMY as working with ENMD on angiostatin.
- ENMD's November 28, 1997 press release said the two firms had forged a strategic partnership in December 1995.

Table II reports BMY's return, excess return, volume, and relative volume on four relevant dates.

- **November 28, 1997:** return **0.40%**, excess return **0.04%**, volume **1.607 million shares**, relative volume **0.85**.
- **May 4, 1998:** return **3.12%**, excess return **2.98%**, volume **8.671 million shares**, relative volume **1.57**.
- **November 12, 1998:** return **-1.29%**, excess return **-1.14%**, volume **1.805 million shares**, relative volume **0.27**.
- **February 10, 1999:** return **-0.20%**, excess return **-0.55%**, volume **5.825 million shares**, relative volume **0.81**.

Interpretation.

- Only **May 4** looks truly unusual for BMY.
- The 3.12% return seems small next to ENMD's jump, but for a firm of BMY's size it implied roughly **\$3.3 billion** in market value creation.
- That is more than four times the combined dollar appreciation of ENMD plus the seven biotech winners on May 4.
- ABI Inform does not reveal other significant BMY-specific news on May 2–4, so the May 4 reaction is plausibly tied to the publicity shock surrounding ENMD.

3.5 What the paper is really showing

A useful note-level decomposition is

$$\Delta P = \underbrace{\Delta P^{\text{new information}}}_{\text{November 1997, November 1998}} + \underbrace{\Delta P^{\text{publicity / salience}}}_{\text{May 1998 nonevent}}. \quad (11)$$

- This equation is **not** in the paper as a formal model; it is the paper’s economic interpretation in compact form.
- The case study suggests that the publicity term can be very large, very persistent, and contagious across related securities.

4 Identification and instruments (what makes the estimates credible)

4.1 There are no instruments in the econometric sense

- The paper does **not** use an instrumental-variables design.
- Identification comes from a **chronology-based comparison**: the same substance was public in November 1997, then repackaged with much greater salience in May 1998.
- The later *Wall Street Journal* failure-to-replicate story gives a third benchmark, showing how the market responded to clearly new bad information.

4.2 Why the May 3 article is a clean “nonevent”

- The key scientific article had already appeared in *Nature* on November 27, 1997.
- The *New York Times* itself had already covered the story on page A28 that same day.
- ENMD had already issued a press release on November 28, 1997 describing the news and its licensing rights.
- Other national media outlets had also covered the development in late November 1997.
- A week after the May 3 article, a *New York Times* piece by Abelson explicitly noted that professional investors were already familiar with the research and had reflected it in the stock’s pre-runup price.

Interpretation.

- This makes it unusually hard to argue that May 3 delivered materially new information.
- What changed was visibility, prominence, and framing — not the underlying scientific fact.

4.3 Why cross-stock evidence strengthens the argument

- If May 4 had affected only ENMD, one could still argue that some subtle firm-specific reinterpretation took place.
- But the reaction spreads to unrelated or only loosely connected biotech firms, including seven firms with no obvious contemporaneous news coverage.
- It also reaches BMJ, a giant pharmaceutical company, on a day when the paper finds no other significant BMJ-specific news.

Interpretation.

- The sector-wide reaction looks more like contagious attention and thematic excitement than careful updating about each firm’s fundamentals.

4.4 What makes the case credible

- **Event timing is transparent.** The paper can line up publication dates, price reactions, and follow-up reporting precisely.
- **The event is unusually clean.** There was an earlier scientific disclosure, a later front-page publicity shock, and then an even later genuinely negative news release.
- **The paper checks permanence.** It is not just a one-day spike story; a meaningful share of the May 4 increase remained for weeks.
- **The paper checks spillovers.** Both small biotech firms and BMY reacted on the nonevent day.

4.5 Main limitations

- **Single-case design.** The paper is convincing because the case is unusually clean, but that also limits generalizability.
- **No structural decomposition.** The paper cannot quantify exactly how much of the May 4 move reflected pure attention, how much reflected coordination in beliefs, and how much reflected expected future financing benefits.
- **No investor-level data.** We do not observe who traded and why. The paper therefore cannot distinguish cleanly between naive noise traders, media-driven retail investors, and sophisticated traders front-running attention.
- **Residual omitted-news concern can never be reduced to zero.** The authors work hard to rule out obvious alternative news, but a skeptic can always imagine subtle reinterpretations or private information becoming focal on May 4.

4.6 Relation to the literature

- This paper is an early and especially vivid media-attention paper in finance.
- Later work on media and investor attention often uses Huberman and Regev as a motivating example because it cleanly separates *information content* from *publicity intensity*.
- Conceptually, it sits at the intersection of efficient-markets debates, investor attention, salience, and contagion across related securities.

5 Tables and figures

5.1 Figure 1: ENMD price and trading volume, October 1997 to December 1998

Read.

- The graph places three dates on the same price-volume path: November 28, 1997; May 4, 1998; and November 12, 1998.
- The visual impression is immediate: the May 4 spike dwarfs the November 1997 jump.

What to remember.

- Real scientific news in late November 1997 produced a noticeable but moderate move.
- The May 4 nonevent produced the giant jump.
- The later negative replication news produced a sharp drop, but not a full reversal of the May 4 revaluation.

Why it matters.

- Figure 1 is the paper's cleanest visual summary: if one only looked at the price path, one would naturally think the biggest information event happened in May 1998. The whole point of the paper is that it did not.

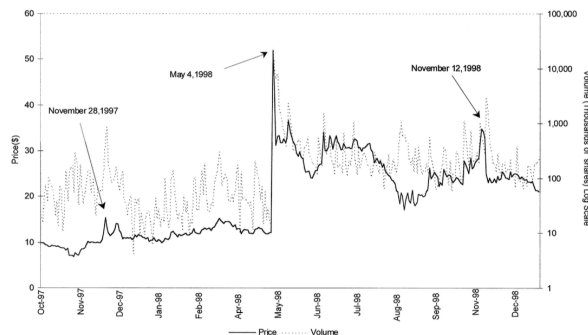


Figure 1. ENMD closing prices and trading volume, October 1, 1997, to December 30, 1998.

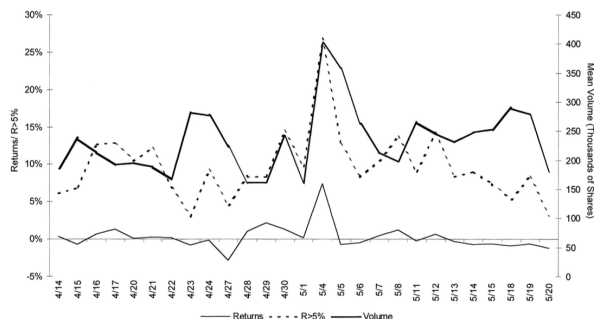


Figure 2. Biotech Index Performance around May 4. $R > 5\%$ is the fraction of stocks with daily return exceeding 5 percent.

5.2 Figure 2: Biotechnology index performance around May 4

Read.

- The figure plots the equally weighted biotech index return around May 4, average trading volume in index members, and the fraction of stocks with daily returns above 5%.

What to remember.

- There is a visible spike in returns, volume, and the share of big winners right around May 4.
- This is not a story of one eccentric stock.

Why it matters.

- Figure 2 shows that the nonevent shock spread through a thematic sector, which is exactly why the title begins with “Contagious Speculation.”

5.3 Table I: frequency of extreme returns and comovement in the biotech index

Read.

- The table counts how often very high returns and clusters of high returns occurred among Biotechnology Index members in 1996–1997.

Key facts.

- Over roughly **56,800 firm-days**, there was only **one** day on which a stock more than doubled.
- Only **twice** did four firms return more than 25% on the same day, and those clustered returns were far smaller than the May 4 winners.

Why it matters.

- Table I quantifies just how statistically strange the May 4 biotech reaction was.
- The paper is not cherry-picking a merely “interesting” day. It is highlighting an event that looks extreme even against a naturally volatile biotech background.

Table I
Frequency of Extreme Returns and Comovements of Returns of Members of the Nasdaq Combined Biotechnology Index Companies
Number of firm-days on which very high or very low daily returns (R) were realized for members of the Nasdaq Combined Biotechnology Index in 1996 and 1997. Total number of firm-days in the sample is 56,800, distributed over 507 trading days. For example, in 1996 and 1997, there was only one day when three members of the Nasdaq Combined Biotechnology Index experienced a daily return above 25 percent.

No. Firms per Day	$R > 100\%$	$R > 50\%$	$R \leq 50\%$	$R > 25\%$	$R \leq 25\%$
1	1	12	4	88	41
2				7	3
3				1	
4				2	

Table II
Daily Returns, Excess Returns, Trading Volume, and Relative Trading Volume for Bristol-Myers Squibb (BMY)

Excess return is the return of BMY in excess of that of the NYSE. Fraction of 1996 and 1997 excess returns higher is the fraction of the 507 1996 and 1997 daily excess returns that were higher than BMY on that day. Fraction of 1996–1997 volume ratios higher is the fraction of the 507 1996 and 1997 daily (BMY volume)/(NYSE volume) ratios that were higher than the similar ratio on that day.

Date	Return	Excess Return	Fraction of 1996–1997 Excess Returns Higher	Volume (Thousands of Shares)	(BMY Volume)/(NYSE Volume) (%)	Fraction of 1996–1997 Volume Ratios Higher
11/28/97	0.40%	0.04%	0.774	1,607	0.85	0.502
05/04/98	3.12%	2.98%	0.044	8,671	1.57	0.024
11/12/98	−1.29%	−1.14%	0.367	1,805	0.27	1.000
02/10/99	−0.20%	−0.55%	0.680	5,825	0.81	0.559

5.4 Table II: Bristol-Myers Squibb on four dates

Read.

- The table reports BMY’s return, excess return, share volume, and relative volume on four ENMD-related dates.

What to remember.

- November 28, 1997: not unusual for BMY.
- November 12, 1998: not unusual for BMY.
- February 10, 1999: also not unusual for BMY.
- **May 4, 1998: unusual for BMY.**

Why it matters.

- This table rules out the idea that all ENMD-related dates generated broad pharmaceutical repricing. The unusual BMY response is tied specifically to the high-salience May 3 publicity event.

6 Short summary

- **Method.** A forensic event-study comparison of one true-news date, one no-new-news publicity date, and one later bad-news date.
- **Main conceptual result.** Publicity can move prices strongly and persistently even when the underlying information has already been public for months.
- **Main firm-level fact.** ENMD rose about **330%** on May 4, 1998 after a highly prominent *New York Times* story that repeated old information.
- **Main cross-stock fact.** The biotech sector also jumped: the equally weighted index rose **7.5%**, and seven stocks without obvious new firm-specific news rose more than **25%**.
- **Main interpretation.** The stock market appears to respond not only to new information, but also to salience, attention, and contagious optimism.
- **Why the paper matters.** It is one of the clearest examples that prices can move on what is essentially a media-amplified nonevent.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a highly salient media event can trigger a large and partly permanent stock-price reaction even when the article conveys no genuinely new information. It also shows that such enthusiasm can spill over to related firms, producing sectoral contagion.

7.2 Economic intuition

The most natural intuition is that attention and publicity change *who notices*, *who talks*, and *who trades*. The May 3 story did not create new scientific knowledge, but it changed the social prominence of the story, framed it in unusually optimistic language, and appears to have coordinated a burst of speculative attention. Once that attention shock occurred, prices adjusted sharply, and part of the move persisted because beliefs, financing conditions, or both did not immediately snap back.

7.3 Takeaway

If one wants to understand asset prices, it is not enough to ask: *What information arrived?* One must also ask: *How was the information packaged, who saw it, and how much attention did it attract?* Huberman and Regev's answer is that these distinctions can matter enormously — enough to make a nonevent look like the defining event in a stock's history.